QUEEN MARY ML SOCIETY · KAGGLE PLAYGROUND S6E11



Predicting Credit Card Fraud

November's competition: find the 1 in 60.

OPEN 1 Nov 2026 CLOSE 30 Nov METRIC ROC AUC

What the Playground Series is

01

A new one every month

Each competition is a fresh tabular dataset that runs for a few weeks.

02

Synthetic data

Generated from a real dataset, so the features have names but the test labels stay hidden.

03

Low stakes

No medals or points. The top 3 win Kaggle merch. Made for practising.

The task: is this transaction fraud?

For each transaction in the test set, predict the **probability** that is_fraud = 1.

700k	300k	24	0
training rows	test rows to predict	features	missing values



THE ONE THING TO KNOW

1.7%

of transactions are fraud. That's about 1 in 60.

A model that always says "not fraud" scores 98.3% accuracy and catches nothing. That's why accuracy is the wrong metric here.

What's in the data

The transaction

amount_usd · time_of_day_hour · day_of_week · merchant_category · channel · device_type

Behaviour

hours_since_last_txn · txn_count_last_24h · velocity_score · distance_from_home_km · cvv_retry_count

Customer and card

customer_age · card_age_months · account_balance_usd · card_type · auth_method · prior_disputes · merchant_risk_score

Risk flags (true/false)

used_vpn · ip_country_mismatch · billing_shipping_mismatch · is_new_merchant · is_foreign_transaction · is_ai_generated_scam_attempt

Where the signal is

Fraud rate when the condition holds. Baseline is 1.7%.

Condition	Fraud rate	vs baseline
ip_country_mismatch = True	16.5%	~10×
is_ai_generated_scam_attempt = True	15.2%	~9×
merchant_category = Gift Cards	7.3%	~4×
auth_method = No Authentication	6.4%	~4×
auth_method = Biometric	0.5%	÷3
channel (any value)	1.6–1.7%	no signal

How you’re scored: ROC AUC

Does fraud score higher?

Compare a fraud and a genuine transaction. Did the fraud get the higher score? AUC is how often the answer is yes.

0.5 to 1.0

0.5 is a coin flip and 1.0 is perfect. Class imbalance doesn't inflate it the way it inflates accuracy.

So: submit probabilities

```
predict_proba(X)[:, 1]
not predict(X)
```

Submitting and deadlines

submission.csv

```
id,is_fraud
699960,0.03
699961,0.91
699962,0.002
… 299,984 rows
```

Timeline

OPENED

1 November 2026

CLOSES

30 November

Join and accept the rules to see the data. The public leaderboard scores only part of the test set, so trust your CV score more.

The starter notebook

1. Load the data and check the fraud rate
2. Find the signal: flags, categories, single-feature AUC
3. LightGBM with 5-fold StratifiedKFold
4. Probabilities vs labels comparison
5. Write submission.csv

CV AUC TO BEAT

0.9456

No feature engineering. Runs in a few minutes on Kaggle.

How to climb the leaderboard

THIS WEEK

Get one submission in

Join, run the starter notebook, submit. Being on the board beats being perfect.

NEXT

Feature engineering

Amount ÷ balance, a count of risk flags per row, interactions between flags.

THEN

More data, more models

Add the original dataset. Try CatBoost, XGBoost or AutoGluon, and average them.

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Get the notebook

FIRST TIME · CLONE THE REPO

```
git clone https://github.com/Queen-Mary-Machine-Learning-Society/QMML.git
```

ALREADY CLONED · GET THE LATEST

```
cd QMML
git pull
```

This month's starter notebook and slides are in 2026-2027/kaggle-sessions